Aryaman Arora

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About Me

Computer Science student with a strong foundation in **systems programming**, **low-level software development**, and **machine learning**. Proficient in **C**, **C++**, and **Python**, with hands-on experience developing a **FUSE-based file system**, **key-value database system API**, and implementing **deep learning models** for sentiment analysis. Demonstrated ability to tackle complex technical challenges, optimize system performance, and collaborate effectively in team environments.

Education

University of Toronto, St. George — Toronto, ON
Honors Bachelor of Science, Computer Science — *Sep. 2021 – April 2025*

Technical Skills and Certifications

Languages: Java, Python, C, C++, SQL, HTML, CSS, JavaScript, GDScript/C#, Assembly, LaTeX, XML.
Frameworks: TensorFlow, Keras, PyTorch, Pytest, Django, Flask, React, Node.js, Vite, Vue.
Libraries and Developer Tools: Sklearn, SciPy, pandas, pickle, boto3, psycopg2, nltk, Fusion 360, Gazebo, ROS, RViz.
Certifications: Coursera Deep Learning Specialization (Neural Networks, CNNs, Sequence Models, etc.).

Experience

CTO — May. 2024 – Present
InkTank — *Toronto, ON*

- Executed the end-to-end development of InkTank's website using **Vite** and **Vue.js**, achieving a **50% reduction** in page load times and providing a seamless platform for over **500+ tattoo artists and clients**.
- Worked closely with cross-functional teams to integrate features, leading to a **20% increase in overall website functionality** and ensuring user satisfaction while shaping InkTank's technical strategy.

Software/Robotics Intern — Jun. 2024 – Aug. 2024
Evodyne Robotics Academy — *Mountain View, CA*

- Recreated the Evodog robotic model entirely in code using **CAD modeling**, mastering new tools such as **ROS 2** and **Gazebo** within **5 weeks** to enable virtual simulations that reduced resource usage by over **50%** and saved **100+** hours of testing time through accelerated design iterations.
- Applied **3D meshes** and physics simulations, improving design accuracy by **25%** and enabling quicker iterations.
- Utilized **ROS 2** for robot control and **Ignition Gazebo** for realistic physics simulations, improving testing efficiency and enabling quicker adjustments to the robotic model.

Projects

Key-Value Database System API | *C++, Python, SQL* — September 2024 – December 2024

- Collaborated in a team of 3 to design and implement a scalable key-value database system, incorporating **Memtables, SSTs, Static B+-Trees, Buffer Pools, LSM-Trees, and Bloom Filters**, emphasizing **memory management** and **performance optimization** to achieve a **30% improvement in query efficiency** and support datasets beyond **10GB**.
- Developed a **user-friendly API** for seamless **interaction with the database via terminal commands**, supporting essential operations like **put, get, scan, and delete**.
- Conducted comprehensive **testing and performance benchmarks** on datasets of **over 100,000 key-value pairs**, demonstrating **sub-millisecond query latencies** and co-authored a **technical report** detailing the system's architecture, implementation, and instructions for deployment.

mhapy Sentiment Analysis Model | *Python, Flask, nltk, TensorFlow, Keras* — September 2023 – December 2023

- Developed a sentiment analysis API for mhapy using **TensorFlow** and **Keras**, leveraging **natural language processing (NLP)** techniques to analyze over **10,000** pieces of user-generated content for mental health trends.
- Built and integrated the **Flask**-based backend with robust RESTful API endpoints, ensuring secure and efficient data handling for real-time analysis, achieving **85%** accuracy in sentiment detection.

FUSE File System | *C* — November 2023 – December 2023

- Developed a FUSE-based version of the Very Simple File System (VSFS), incorporating **low-level programming techniques** to implement file operations like creation, deletion, reading, and resizing. Optimized **concurrency** handling and memory allocation, demonstrating expertise in **file system architecture** and **system-level programming**.
- Managed disk formatting and block allocation techniques, utilizing bitmaps for inode and block allocation, improving memory efficiency by over **30%**.

MarkUs Database Analysis | *Python, SQL* — October 2023 – November 2023

- Designed and executed complex **SQL** queries to extract and analyze data from the MarkUs database, improving data accessibility and reporting efficiency by **35%**.
- Implemented **Python** scripts using the **psycopg2** library, automating data extraction and processing for large datasets, reducing manual processing time by **50%**.